Infrastructure as Code

Software Architecture

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March 16, 2026

- Do a quick poll.
- Who has heard the term IaC before this course?
- Who has used IaC before this course?

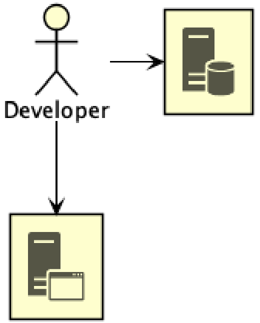
Infrastructure as Code

How did we get here?

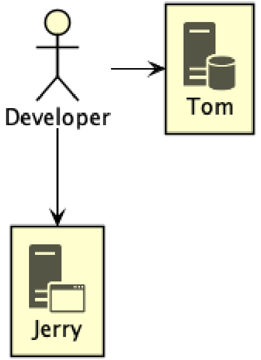
Pre-2000

The *Iron Age*

Iron Age

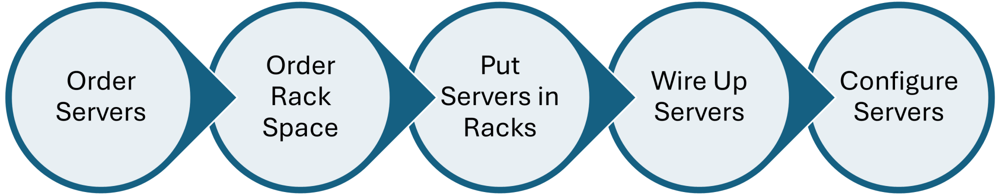


Iron Age



Developer only had a few machines — so few the machines often got fun names.

Scaling

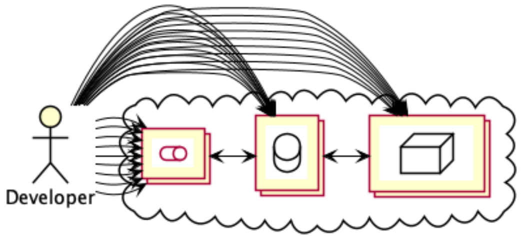


I used to have scheduling examples for software projects, where ordering servers was on the critical path, and had to happen before software development started.

Introducing...

The *Cloud Age*

The Cloud Age



- Things got complicated quickly
  - We need more hardware and it is easier to provision
- Largely thanks to virtualisation
  - No physical activity to spin up a new machine

When faced with complexity

Automate it!

- We have too much to manage to do it manually
- We're about to start enumerating automation techniques

The larger story

Server Config Config Management

The larger story

Server Config	Config Management
Application Config	Config Files

The larger story

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Application Config	Config Files
Provisioning	Infrastructure Code

The larger story

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The larger story

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Database Administration	Schema Migration
Specifications	Behaviour Driven Development

Definition 0. Infrastructure Code

Code that provisions and manages *infrastructure resources*.

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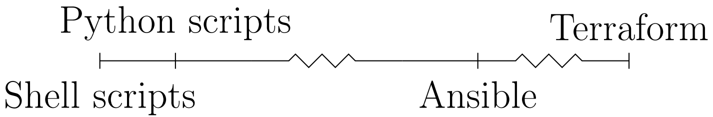
Definition 0. Infrastructure Resources

Compute resources, networking resources, and storage resources.

Infrastructure Code



IC often thought of as the right-hand side, but includes all

Shell Scripts

```
#!/bin/bash

SG=$(aws ec2 create-security-group ...)

aws ec2 authorize-security-group-ingress --group-id "$SG"

INST=$(aws ec2 run-instances --security-group-ids "$SG" \
        --instance-type t2.micro)
```

Using AWS CLI to create EC2 access like in the practical

Python

```python
import boto3

def create_instance():
    ec2_client = boto3.client("ec2", region_name="us-east-1")
    response = ec2.create_security_group(...)
    security_group_id = response['GroupId']

    data = ec2.authorize_security_group_ingress(...)

    instance = ec2_client.run_instances(
        SecurityGroups=[security_group_id],
        InstanceType="t2.micro",
        ...
    )
```

Using AWS Python library (boto3)

Terraform

```
resource "aws_instance" "hextris-server" {
    instance_type = "t2.micro"
    security_groups = [aws_security_group.hextris-server.name]
    ...
}

resource "aws_security_group" "hextris-server" {
    ingress {
        from_port = 80
        to_port = 80
        ...
    }
    ...
}
```

Finally, Terraform

Question

Notice anything different?

- Prompting for declarative
- Might notice verbosity

The main difference

Imperative vs. Declarative

- *Imperative* – Describe the steps to take to deploy the infrastructure
- *Declarative* – Describe the desired infrastructure
- IC is heading towards a more *declarative* paradigm

Declarative IaC

- Define your *desired* infrastructure state
  - As code
- Engine interprets difference between the *desired* and *actual* state
  - Modifying infrastructure to deliver *desired* state

Infrastructure Code

- Provisions and manages *infrastructure resources*

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- Only one part of the movement to *automate* the complexities of development

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- Ranges from simple shell scripts up to...?

Infrastructure Code

- Provisions and manages *infrastructure resources*
- Only one part of the movement to *automate* the complexities of development
- Ranges from simple shell scripts up to. . . ?
- Tendency to be *declarative*

Summarising what we've already covered

Typo?

Infrastructure Code $\neq$ Infrastructure *as* Code

- Mention that this distinction is ours
- Real world unfortunately mixes the two

Definition 0. Infrastructure as Code

Following the same *good coding practices* to manage Infrastructure Code as standard code.

Warning!

Infrastructure as Code still *early* and quite *bad*

- Code reuse is low
- Importing existing resources is non-trivial
- Refactoring is painful
- State management can be tricky

Question

What are *good coding practices*?

Ask the class

Good Coding Practice #1

Everything as Code

A practice we do but barely discuss in ‘regular’ programming because it doesn’t make sense not to do it

```
#!/bin/bash

./download-dependencies
./build-resources
cp -r output/* artifacts/
```

```bash
#!/bin/bash

./download-dependencies
./build-resources
cp -r output/* artifacts/
```

```
$ cp: directory artifacts does not exist
```

An example of relying on external state in ‘regular’ programming

```
resource "aws_instance" "hextris-server" {
    instance_type = "t2.micro"
    security_groups = ["sg-6400"]
    ...
}
```

Draw a parallel to the bash example and this, which relies on ‘sg-6400’ existing

```
resource "aws_instance" "hextris-server" {
    instance_type = "t2.micro"
    security_groups = [aws_security_group.hextris-server.name]
    ...
}

resource "aws_security_group" "hextris-server" {
    ingress {
        from_port = 80
        to_port = 80
        ...
    }
    ...
}
```

The better approach

Everything as code avoids

Configuration drift

Configuration drift creates

Snowflakes

- *Snowflakes* magical machines that ‘just work’ and everyone is afraid to touch
- Snowflake because they’re unique and easy to break, because no one knows how it works

Benefits

1. Reproducible

Good Coding Practice #2

Version Control

Benefits

1. Restorable
2. Accountable

Good Coding Practice #3

Automation

Benefits

1. Consistent

Automatically applying or checking IC is in sync means the main branch is consistent with reality

Good Coding Practice #4

Code Reuse

Benefits

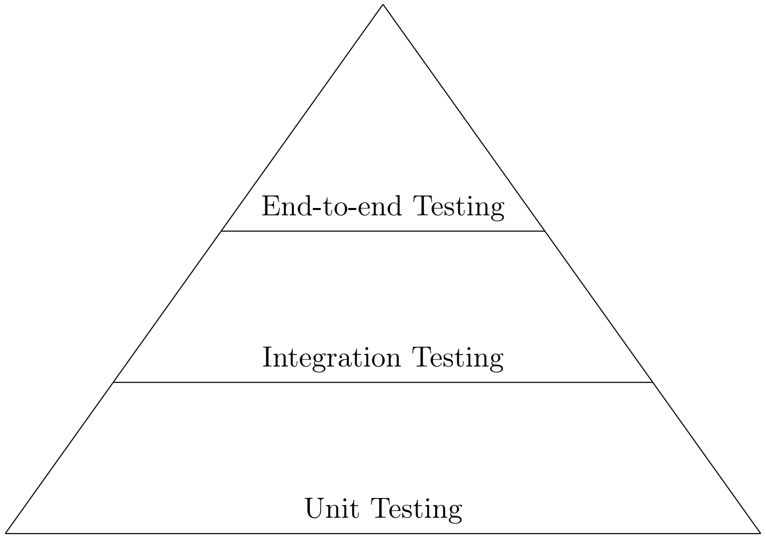
1. Better[1] code
2. Less work
3. Only one place to update (or verify)

[1] generally

Good Coding Practice #5

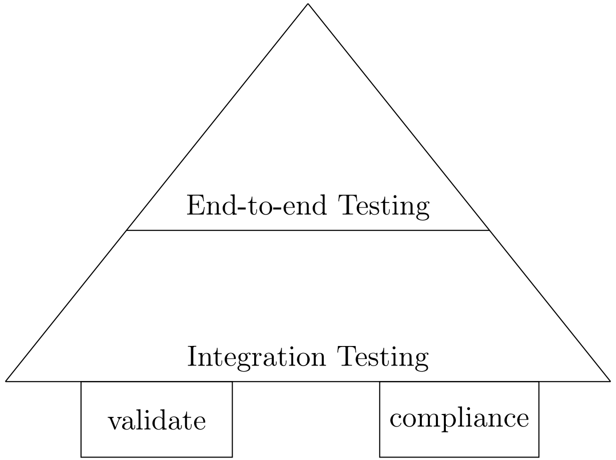
Testing

Test Pyramid



- Traditional test pyramid
- Unit testing relies on isolated testing
- But…isolated testing doesn't make *much* sense for IaC

IaC Test Pyramid



```go
func TestTerraformAwsInstance(t *testing.T) {
    terraformOptions := terraform.WithDefault(t, &terraform.Options{
        TerraformDir: "../week03/",
    })

    defer terraform.Destroy(t, terraformOptions)
    terraform.InitAndApply(t, terraformOptions)

    publicIp := terraform.Output(t, terraformOptions, "public_ip")
    url := fmt.Sprintf("http://%s:8080", publicIp)

    http_helper.HttpGetWithCustomValidation(t, url, nil, 200,
        func(code, resp) { code == 200 &&
                            strings.Contains(resp, "hextris")})
}
```

Validation example

```
Feature: Define AWS Security Groups

Scenario: Only selected ports should be publicly open
    Given I have AWS Security Group defined
    When it contains ingress
    Then it must only have tcp protocol and port 22,443 for 0.0.0.0/0
```

Compliance testing example

Benefits

1. Trust

Week 4 Prac

Learn how to use Terraform to write IaC and deploy resources on AWS.